

2(a)(i)	39 m s^{-1}	A1
2(a)(ii)	tangent line to curve drawn on Fig. 2.1	C1
	$a = \text{gradient of tangent line} = \Delta v / \Delta t$ e.g. $= (44 - 26) / (18 - 0)$ $0.9 \leq a \leq 1.1 \text{ m s}^{-2}$	A1
2(b)	$(\Sigma)F = 68 \times 9.81 - 1800$ $(= -1133 \text{ N})$	C1
	$a = (\Sigma)F / m = -1133 / 68$ $= (-)17 \text{ m s}^{-2}$	A1
	upwards	B1
2c(i)	drag force decreases (as speed decreases)	B1
	(as speed decreases) resultant force decreases so (magnitude of) acceleration decreases (to zero)	B1
2(c)(ii)	$(\Delta)p = m\Delta v$ or $(\Delta)p = m(v - u)$	C1
	$= 68 \times (5.7 - 39)$ $= (-)2300 \text{ N s}$	A1